Experiment 6: Word Sense Disambiguation (WSD)

Aim

To implement Word Sense Disambiguation methods and identify the correct meaning of ambiguous words based on context.

Theory

Word Sense Disambiguation (WSD) is a **Natural Language Processing (NLP)** task used to determine the correct meaning of a word that has multiple meanings depending on context.

Example:

- "bank" → river side
- "bank" → financial institution

Methods of WSD

- **Knowledge-Based (Lesk Algorithm):**
  Uses dictionary meanings and finds overlap with context. Simple but less accurate.
- **Supervised Learning:**
  Uses labeled data and algorithms like Naive Bayes. More accurate but needs training data.
- **Unsupervised Learning:**
  Groups similar contexts using clustering. No labeled data required but less reliable.
- **Deep Learning:**
  Uses models like LSTM/BERT for better contextual understanding. Highly accurate but computationally expensive.

Conclusion

WSD helps machines understand the correct meaning of words in context. In this experiment, it is observed that simple methods like Lesk are easy to use but less accurate, while supervised and deep learning methods provide better results. The choice of method depends on data availability and required accuracy.